MINGWEI ZHANG

New York, NY | 347-532-0885 | mingw.zhang123@gmail.com
linkedin.com/in/mingwei-zhang1 | github.com/mingwzhang | mingwzhang.github.io/portfolio

EDUCATION

Queens College, **CUNY**, Queens, NY
Master of Arts in Computer Science — Expected Dec 2026

- **Relevant Coursework**: Algorithms, **Machine Learning**, Distributed Systems, Database Systems, Operating Systems, Computer Architecture, Theory of Computation, Programming Languages, **Video Game Design**

Stony Brook University, SUNY, Stony Brook, NY
Bachelor of Science in Information Systems and Applied Mathematics & Statistics — Aug 2021

- **Relevant Coursework**: **Data Structures**, Linear Algebra, Database Systems, Computer Security, Computer Communications, Design of Computer Games, **Probability & Statistics**

Certifications🔗

- Full Stack Java Development, CSS, HTML, JavaScript

PROJECTS

Top 5-8% Finishes: Creator-Hosted Game Jams - Brackeys Jam 2026.1, Juniper Dev Jam, Brackeys Jam 2026.2
Competitive Game Jam Projects (itch.io) — Jan 2026 - Sep 2026

- Placed in the **top 5-8%** across large game jams hosted by game-development content creators **Brackeys** and **Juniper Dev**, competing in jams with **1,422-3,502 submissions** and **8,948-17,500 participants**
- Served as **Lead Game Developer, Gameplay Programmer, and Technical Artist**, leading **end-to-end gameplay development**, writing design documentation, and building **AI, combat, UI, player input, animation, and state management systems**
- Managed the **art/audio asset pipeline**, cross-discipline integration, **playtesting, debugging, and iteration**, and expanded one project into a **paid itch.io release**

Machine Learning Project: Semantic Segmentation - Semantic Segmentation on PASCAL VOC 2012 (report & code)🔗
Queens College, CSCI 780 · Fall 2025 — Nov 2025

- Developed and evaluated **U-Net** and **ResNet-50 segmentation models** in **PyTorch** on **PASCAL VOC 2012**, **achieving a 5-15% improvement in mIoU** through transfer learning
- Built an **end-to-end training pipeline** with **data preprocessing**, **augmentation**, and **class-weighted loss**
- Evaluated models using **pixel accuracy** and **mean IoU (mIoU)**, highlighted performance gains from **transfer learning**

Personal Portfolio Website - Portfolio🔗
Self-Developed — Feb 2025

- Built a **mobile-responsive portfolio** using **React.js**, **CSS**, and **JavaScript**
- Integrated **videos**, **mini-game demos**, **animations**, **interactive effects**, and a **contact form** to improve user engagement

Hackathon Prize Winner: Hack Knight Spring 2025 - Banking Simulator🔗
Queens College Hack Knight — Mar 2025

- Built a **gamified banking education app** in a 4-person team using **Next.js, Tailwind CSS**, **JavaScript**, and the **Nessie API** to simulate banking transactions, deposits, and withdrawals
- Won "Funniest Hack" among 46 teams for **innovation**, judged by engineers from **Bloomberg**, **Capital One**, and **Citi**

WORK EXPERIENCE

Game Design Internship (Remote), Scottsdale, Arizona - **Project IOTA**🔗**, Ash: Protector of Ignicity**🔗**, FireFighter Blaze**🔗
MoonRift Entertainment, LLC — Sep 2021 - Dec 2022

- Designed **2 third-person** and **1 first-person games** using **Unity** and **Unreal Engine 5** in interdisciplinary teams of 2-5
- Supported **UI design**, **3D modeling, animation** (**Blender**), and **debugging** (**C# and Blueprints**) for team projects
- Managed **version control** with **GitHub** and **Unity Collaborate**, ensured smooth code and data transfer across **100+ commits**
- Implemented 8+ game mechanics using **C#** (interaction systems, puzzles, missions, AI), **accelerated project completion by 30%**

Contract Game Developer (Remote), Davis, California - **Marine Biology**🔗**, Celestial Journey** 🔗**, Music Hall**🔗**, Maze Puzzle**🔗
AlterLearning Educational Platform, Inc — Mar 2022 - Sep 2022

- Developed **4 VR educational games** using **Unity**, collaborating with teams of 5+
- Managed **version control** with **GitHub** and task management on **Trello**, improving team alignment and project timelines
- Created 5+ prototype models and 3D assets using **Blender**, **improved iteration cycles by 25%** and project timelines by **2 weeks**
- **Increased user interactions by 10%** with 10+ game mechanics (inventory systems, VR interactions, mini-games) using **C#**

TECHNICAL SKILLS

- **Technical: Python**, JavaScript, **C#**, C++, **Java**, R, CLISP, **MySQL**, MS Access, Oracle, Deep learning, convolutional neural networks (CNNs), transfer learning, model evaluation (IoU, mIoU), Git, **Unity**, Unreal Engine 5, **Godot**, React, Tailwind CSS, Next.js, Spring, JSP, Servlet, PyTorch
- **Soft: Game Design, Data Analysis, Design Documentation**